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The Essentials

Introduction and Motivation

The **k-means** algorithm is a method for **estimating discrete latent factors** in an **unsupervised** clustering context.

The Fundamental Problem

Imagine this: You're given 1,000 unlabeled animal photos and asked: "How many different types of animals are there, and which photos go together?" This is exactly the clustering problem!

K-means answers this question by making **a simplifying assumption**: each type of animal can be represented by an **average image** (prototype), and each photo is close to one of these prototypes.

Key Concept: Clustering is **unsupervised**: no labels, no teacher to say "this is a cat." The algorithm must discover the structure by itself.

Real Data Problem

Fundamental Observation:

Data generally does **not** come from a simple Gaussian process (variations around a mean prototype).

Characteristics of Real Data:

- Data distribution typically shows **non-uniformity**
- Presence of **regions of higher density**

Definition of Clustering:

Clustering is an **unsupervised** method that aims to discover **coherent groups** of data, corresponding to **density peaks** in the data.

Geometric Intuition: The Musical Chairs Game

Practical Analogy

Imagine a game of musical chairs with K chairs (clusters) and N players (data points). When the music stops:

- Each player runs to the nearest chair **(assignment)**
- Each chair is moved to the center of the players surrounding it **(update)**
- Repeat until the chairs no longer move

This is exactly k-means!

Essential Idea: K-means alternates between two simple questions: "where does each point go?" and "where are the centers?" This alternation guarantees improvement at each step.

Overview of Clustering Methods

Main Methods

There are many clustering methods, including:

Hierarchical Methods:

- Hierarchical Agglomerative Clustering (HAC)

Partitioning Methods:

- **k-means** (Lloyd's algorithm)
- k-medoids

Spectral Methods:

- Spectral clustering

Other Approaches:

- Community detection
- High-dimensional clustering
- Self-Organizing Maps (SOM)
- Neural Gas

Historical Note: Research on clustering has been active for at least 50 years.

Hierarchical Agglomerative Clustering

Algorithm Principle

Iterative Process:

- **Initialization:** Each data point is a cluster
- **Search:** Find the closest pair of clusters
- **Merge:** Merge these two clusters
- **Iteration:** Repeat until completion

Result: Produces a **dendrogram** of the data.

Distances Between Sets (Clusters)

Possible Metrics:

- **Distance between centers** (centroid linkage)
- **Maximum distance** (complete linkage): $\max_{x \in C_1, y \in C_2} d(x, y)$
- **Minimum distance** (single linkage): $\min_{x \in C_1, y \in C_2} d(x, y)$

Limitation: The number of clusters is **unknown a priori**.

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k-means Strategy

Fundamental Assumptions

The **k-means** clustering algorithm assumes:

- A **normed metric space** (Euclidean) on the data
- An **unsupervised** assignment of data to clusters
- A **hard-assignment** algorithm: assignment is **binary**, each data point is assigned to **one and only one** cluster

Formal Model

Data:

Let $X = \{x_i\}_{i \in [[N]]} \subset \Omega \subseteq \mathbb{R}^D$, and given $K \in \mathbb{N}^*$

Model Variables:

- **Binary assignment variables (latent):**

$$Z = \{z_{ik}\} \quad \text{with} \quad z_{ik} \in \{\text{false}, \text{true}\} \equiv \{0, 1\} \quad \forall i, k$$

- **Cluster representatives:**

$$M = \{\mu_k\}_{k \in [[K]]} \quad \text{with} \quad \mu_k \in \mathbb{R}^D \quad \forall k$$

Complete Parameters:

$$\theta = [M, Z]$$

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k-means Loss Function

Optimization Problem

k-means seeks the following assignment:

$$\hat{\theta} = [\hat{M}, \hat{Z}] = \arg\min_{\theta=[M,Z]} L(\theta, X)$$

Loss Function:

$$L(\theta, X) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \|x_i - \mu_k\|_2^2$$

Interpretation:

- Sum of **squared distances** between each point and its cluster representative
- The smaller this sum, the better the partition

Why squared? Two reasons:

- **Mathematical:** makes the problem easier to solve
- **Geometric:** penalizes more points that are very far away

Problem Complexity

Major Difficulty: Since the assignment is binary, exact optimization is **NP-Hard**.

Practical Solution: Seek an **approximation via coordinate descent**.

Coordinate Descent Algorithm

General Principle

Coordinate descent is an **alternating minimization algorithm**.

Problem:

Let $f : \mathbb{R}^D \to \mathbb{R}$, we seek:

$$x^* = \arg\min_{x \in \mathbb{R}^D} f(x)$$

Mathematical Details: Resolution Strategy

Definition of Partial Functions:
For $d \in [[D]]$, define $f^d : \mathbb{R}^D \times \mathbb{R} \to \mathbb{R}$:

$$f^d(x, y) = f([x(1), \dots, x(d-1), y, x(d+1), \dots, x(D)]^T)$$

Alternating Optimization:
Seek alternately the minimum along each coordinate:

$$x^{(t+1)}(d) = \arg\min_y f^d(x^{(t)}, y)$$

Principle: Fix all variables except one, optimize this variable, then move to the next.

Application to k-means

Alternating Optimization of Z and M

We **alternate the optimization** of Z and M.

Definitions of Partial Losses:

$$L_M(Z) = L(\theta, X)|_{M=M}$$

$$L_Z(M) = L(\theta, X)|_{Z=Z}$$

The Two Magic Equations

Optimization with Respect to M (Representatives):

Mathematical Details: Center Calculation

Derivative Calculation:

$$\frac{\partial L_Z}{\partial \mu_k} = -2 \sum_{i=1}^{N} z_{ik}(x_i - \mu_k)$$

Optimality Condition:
The optimal representative M for a given assignment Z is reached when:

$$\frac{\partial L_Z}{\partial \mu_k} = 0 \quad \Rightarrow \quad \mu_k = \frac{\sum_{i=1}^{N} z_{ik} x_i}{\sum_{i=1}^{N} z_{ik}}$$

Fundamental Conclusion: Cluster representatives are their **centers of mass** (barycenters).

Optimization with Respect to Z (Assignment):

Given a set of cluster representatives M, to minimize $L_M(Z)$:

$$z_{ik} = 1 \quad \text{only when} \quad k = \arg\min_{m} \|x_i - \mu_m\|_2^2$$

(and $z_{ik} = 0$ otherwise)

Conclusion: $L_M(Z)$ is minimized when **each data point is assigned to its nearest cluster representative**.

Complete k-means Algorithm

Detailed Description

Input Data:

- $X = \{x_i\}_{i \in [[N]]}$ with $x_i \in \mathbb{R}^D$
- $K \in \mathbb{N}^*$ (number of clusters)

Algorithm Steps

Step 1: Initialization

Initialize cluster representatives $M^{(0)}$.

Step 2: Assignment (E-step like)

Given cluster representatives $M^{(t)}$, assignment $Z^{(t)}$ associates each data point with its nearest cluster representative:

$$z_{ik}^{(t)} = \begin{cases} 1 & \text{if } k = \arg\min_m \|x_i - \mu_m^{(t)}\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

Step 3: Update (M-step like)

Given assignment $Z^{(t)}$, new cluster representatives $M^{(t+1)}$ are the centers of mass of data points assigned to clusters:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{N} z_{ik}^{(t)} x_i}{\sum_{i=1}^{N} z_{ik}^{(t)}}$$

Step 4: Convergence

Repeat from step 2 until convergence.

Convergence Criteria

Convergence is reached when:

- The centers of mass **do not move much**: $\|\mu_k^{(t+1)} - \mu_k^{(t)}\| < \epsilon$
- Or the assignment is **stable**: $Z^{(t+1)} = Z^{(t)}$

Important Note: Step 2 is similar to an **Expectation** step, and step 3 is similar to a **Maximization** step, considering a hard assignment (reference to the EM algorithm).

Properties of the k-means Algorithm

Theoretical Guarantees

Loss Decrease:

Since it performs **alternating minimization of convex functions**, it guarantees to **decrease the loss at each iteration**:

$$L(M^{(t+1)}, Z^{(t+1)}) \leq L(M^{(t)}, Z^{(t)})$$

Finiteness:

There is a **combinatorial but finite** number of possible assignments → the number of iterations is **(large but) finite**.

Why does it converge? Each step can only improve (or leave unchanged) the criterion. Since there's a finite number of possible assignments, the algorithm must terminate.

Geometric Interpretation

Voronoi Diagram:

Step 2 is equivalent to constructing the **discrete Voronoi diagram** of X with centers M as seeds.

Definition: The Voronoi diagram partitions space into regions, where each region contains all points closer to a given center than to any other center.

Link with the EM Algorithm

Conceptual Comparison:

- **Step 2 (Assignment):** similar to an **Expectation** step (but hard)
- **Step 3 (Update):** similar to a **Maximization** step

Key Difference: k-means uses **hard assignment** while EM uses **soft assignment**.

Sensitivity to Initialization

Problem of Local Optima

Major Limitation: Since exact optimization (optimal assignment) is **NP-Hard**, the k-means algorithm reaches a **local optimum**.

Consequences:

- The quality of the result **varies with initialization**
- Different initializations can yield very different results

The Initialization Dilemma

If all chairs start in the same corner, the game will be very long and yield a poor result. **K-means is very sensitive to initialization!**

Practical Strategies

Randomization:

Randomization can be useful if computation is fast $\rightarrow$ run multiple times with different initializations.

Heuristics:

Otherwise, use **heuristics for better initialization** (like k-means++).

Initialization, Links, and Relationships

k-means++: Smart Initialization

Fundamental Principle

The principle of **k-means++** is to **maximally disperse** the initial cluster representatives $M^{(0)}$ over the data.

Demonstrated Advantages:

- Produces **higher quality** clusters
- **Accelerates convergence** of k-means

Detailed k-means++ Algorithm

Step 1: First Center

Select the first cluster representative μ_1^0 **randomly** from the data X.

Step 2: Distance Calculation

For all unselected data points x_i, calculate:

$$\Delta_i = \|x_i - \mu_m^0\|_2^2$$

the distance between x_i and its nearest representative μ_m^0 among the k representatives already selected.

Step 3: Weighted Sampling

Sample the next representative μ_{k+1}^0 from X with **probability proportional** to the distribution Δ:

$$\mathbb{P}(\text{choosing } x_i) = \frac{\Delta_i}{\sum_j \Delta_j}$$

Step 4: Iteration

Repeat from step 2 until K representatives are chosen.

Algorithm Intuition

Key Idea: Points **far from already selected centers** have a higher chance of being chosen as new centers.

Consequence: Initial centers are well **distributed in the data space**, covering different density regions.

Practical Adoption

Widespread Use: This initialization strategy is used by several Data Science packages: MATLAB™, Python SciKit Learn, R, and other standard libraries.

Practical Tip: Always use k-means++ or run k-means multiple times with different random initializations and keep the best result.

Relationship Between k-means and Gaussian Models

k-means as a Special Case of GMM

Question: Why do we say that k-means considers a **Gaussian model** for clusters?

Answer:

k-means can be seen as a limiting case of GMM (Gaussian Mixture Model) where:

Model Assumptions:

- Each cluster follows a **normal distribution**
- **Spherical and identical** covariance matrices: $\Sigma_k = \sigma^2 I_D$ for all clusters
- Variance $\sigma^2 \to 0$: distributions become spikes

Consequence:

In this limit, the **soft assignment** of GMM becomes a **hard assignment**, and we recover k-means.

Mathematical Details: Probabilistic Formulation

If we consider the probabilistic model:

$$\mathbb{P}(x|\mu_k, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x-\mu_k\|^2}{2\sigma^2}\right)$$

When $\sigma \to 0$, maximizing this probability is equivalent to minimizing $\|x - \mu_k\|^2$, which is exactly the k-means criterion.

k-means is a special case of GMM! When we take a mixture of Gaussians with spherical and identical covariances $\Sigma_k = \sigma^2 I$, and let $\sigma \to 0$, then soft assignment becomes hard, and we recover k-means.

Intrinsic Link with the Euclidean Metric

Why k-means Depends on the Euclidean Metric?

Analysis of the Loss Function:

$$L(\theta, X) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \|x_i - \mu_k\|_2^2$$

The norm $\|\cdot\|_2$ is the **Euclidean norm**: $\|x\|_2^2 = \sum_d x(d)^2$.

Practical Consequences

Scale Sensitivity:

k-means is **sensitive to scale**. Variables with large values dominate the distance.

Solution: Normalize the data before applying k-means:

- Center: subtract the mean
- Scale: divide by the standard deviation

Limitation to Euclidean Spaces:

k-means cannot be directly applied to **non-Euclidean** data (graphs, texts, etc.).

Alternatives: For other metrics, use variants like k-medoids that work with arbitrary distances.

Choosing the Number of Clusters K

Problem

Difficulty: The number of clusters K must be **specified a priori** in k-means.

Selection Methods

Elbow Method:

Plot the loss L as a function of K and look for an "elbow" in the curve.

Silhouette Score:

Measures clustering quality by comparing intra-cluster distance to inter-cluster distance.

For a point i:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where $a(i)$ = average distance to other points in the same cluster, $b(i)$ = average distance to points in the nearest other cluster.

Interpretation: $s(i) \approx 1$ = well classified, ≈ 0 = on the border, ≈ -1 = misclassified.

Gap Statistic:

Compares observed loss to expected loss under a reference distribution.

Information Criteria:

- BIC (Bayesian Information Criterion)
- AIC (Akaike Information Criterion)

Practical Advice: Test several values of K and use multiple criteria.

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k-means Pitfalls and How to Avoid Them

Pitfall 1: Non-Spherical Clusters

Problem: k-means assumes spherical clusters. If they're elongated or oddly shaped, it fails.

Solution:

- Standardize the data (center and scale)
- Use another method (GMM, spectral clustering)
- Transform the data (PCA first)

Pitfall 2: Very Different Cluster Sizes

Problem: k-means tends to create clusters of similar size, even if reality is different.

Solution: Adjust K, or use a method that handles this (GMM).

Pitfall 3: Sensitivity to Outliers

Problem: A very distant point pulls its cluster center toward it.

Solution:

- Detect and remove outliers beforehand
- Use k-medoids (more robust)
- Standardize to reduce influence

Pitfall 4: Result Interpretation

Problem: k-means always finds K clusters, even if the data has no structure.

Solution: Always validate with objective methods (silhouette, gap statistic).

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k-means Variants

k-medoids

Difference: Centers are actual data points (not averages).

Advantage: More robust to outliers, works with any distance.

Mini-batch k-means

Difference: Uses data subsamples at each iteration.

Advantage: Much faster for large datasets.

Fuzzy C-means

Difference: Soft assignment (like GMM) but with a geometric criterion.

Advantage: Border points belong partially to multiple clusters.

Constrained Clustering

Clustering can be **constrained** with:

MUST-LINK Constraints:

Two points must be in the **same cluster**.

CANNOT-LINK Constraints:

Two points must be in **different clusters**.

Practical Application: These constraints allow incorporating **prior knowledge** or **domain constraints** into the clustering process.

In Practice: The Complete Recipe

Step 1: Data Preparation

- **Clean:** handle missing values
- **Standardize:** center and scale each variable
- **Visualize** (if possible in 2D/3D) to get an idea

Step 2: Choosing K

- **Elbow method:** plot inertia vs K
- **Silhouette score:** average over all points
- **Domain knowledge:** sometimes we know how many clusters to look for

Step 3: Execution

- Initialize K centers with k-means++
- Repeat:
  - Assign each point to the nearest center
  - Recalculate centers as averages
  - If centers move little, stop
- Evaluate with silhouette score
- If in doubt, rerun with different initializations

Step 4: Validation

- **Internal:** scores (silhouette, Davies-Bouldin)
- **External:** if you have some labels, compare
- **Manual:** visualization, common sense

Comparison: k-means vs GMM vs Hierarchical Clustering

Criterion	k-means	GMM (with EM)	Hierarchical Clustering
Assignment Type	Hard	Soft	Hierarchical
Cluster Shape	Spherical (same variance)	Elliptical (different covariances)	Flexible
Number of Clusters	Must be specified	Must be specified	Determined a posteriori (dendrogram)
Complexity	$O(N \cdot K \cdot D \cdot T)$	$O(N \cdot K \cdot D^2 \cdot T)$	$O(N^2 \log N)$ or $O(N^3)$
Advantages	Fast, simple	Probabilistic model, varied cluster shapes	No need to specify K, visualization
Disadvantages	Sensitive to initialization, assumes spherical clusters	Slower, more parameters	Expensive for large data, no reassignment

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Synthesis of Key Concepts

Essential Characteristics of k-means

Nature of the Method:

- k-means is part of the **unsupervised** family of data modeling techniques
- One of the **most popular** clustering techniques

Assignment:

- k-means performs **hard assignment**
- Each point belongs to **one cluster only**

Optimization:

- **Solid** but **NP-Hard** optimization criterion (loss)
- The k-means algorithm seeks an **approximate solution** via coordinate descent (alternating minimization)

Convergence:

- The loss **is not convex** → technique **sensitive to initialization**
- k-means++ is a **prior heuristic** for initialization, effective in practice

Extensions:

- Clustering can be **constrained** (MUST-LINK and CANNOT-LINK constraints)

Condensed Cheat Sheet

In 30 seconds for the oral exam:

- k-means = clustering by distance minimization
- Two steps: assignment (points → nearest center) and update (recalculate centers)
- Problems: choice of K, initialization, spherical clusters
- Solutions: k-means++, elbow method, standardization
- Relationship: special case of GMM with $\sigma \to 0$
- Applications: segmentation, grouping, dimensionality reduction

Mathematical Concepts to Master

Algebra and Geometry:

- Euclidean norm and distance
- Center of mass (barycenter)
- Voronoi diagram
- Metric spaces

Optimization:

- Coordinate descent
- Alternating minimization
- Combinatorial optimization
- NP-Hard complexity
- Convergence of iterative algorithms
- Local vs global optima
- Convex functions

Probability and Statistics:

- Multivariate Gaussian distribution
- Maximum likelihood
- Mixture models

Points to Develop:

- Derivation of the loss function
- Calculation of $\frac{\partial L_Z}{\partial \mu_k}$ and optimality condition
- Proof that centers are barycenters
- Proof of monotonic loss decrease
- Relationship between k-means and limiting Gaussian model